Light yield, not wavelength-shifter species, sets the time resolution of RADiCAL sampling-calorimeter modules

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Abstract

We compare the precision-timing performance of four builds of the RADiCAL tungsten/LYSO:Ce sampling electromagnetic calorimeter module that differ primarily in their wavelength-shifter (WLS) capillaries – the common LYSO:Ce scintillator and tungsten absorber are unchanged – exposed to 25–150 GeV electrons at the CERN SPS: a high-light DSB1 (organic WLS) build, a low-light LuAG:Ce (ceramic WLS) build, a mixed DSB1/LuAG:Ce build, and a multi-segment build. The high-gain channels used for timing saturate, clipping at 820 mV; we recover the unclipped rising edge event by event from a linear low-gain copy of each pulse, timing a constant fraction of the low-gain-predicted true peak on an edge a factor ~ 3.7 steeper than the clipped-peak foot, which improves the resolution by a few picoseconds over timing the clipped pulse. Using a microchannel-plate-free differential estimator, $(\mathrm{DW}-\mathrm{UP})/2$, formed from the four upstream and four downstream silicon-photomultiplier readouts, we measure the time resolution of every build and parametrise it as $\sigma_t = a/\sqrt{E} \oplus b$. The stochastic term a scales with the collected scintillation light, more than doubling from the high-light to the low-light build, whereas the floor $b \approx 20$ ps is consistent across the LYSO-based builds and with the value previously reported for DSB1 [NIM A 1068 (2024) 169737]. The time resolution is therefore governed by light yield, not by the wavelength-shifter species: at equal light all materials approach the same floor. The high-light build reaches 25.4 ps for the brightest 150 GeV showers. We discuss light yield as the primary design lever and the material-independent floor as shared shower/detector physics.

Keywords: Fast timing, Sampling calorimeter, LYSO, LuAG:Ce, Wavelength shifter, Silicon photomultiplier, Signal saturation, 4D calorimetry

1. Introduction

Future high-luminosity colliders demand electromagnetic (EM) calorimeters that resolve the time of a shower to a few tens of picoseconds, in addition to its energy, to associate deposits with the correct primary vertex and reject out-of-time pile-up [1]. The RADiCAL R&D programme develops ultra-compact, radiation-hard sampling EM calorimeter modules toward this goal [2].

A first RADiCAL test-beam measurement, on a single high-light build, reported a time resolution of ≈ 27 ps at 150 GeV and parametrised the energy dependence as $\sigma_t = a/\sqrt{E} \oplus b$ with $a = 256$ ps$\sqrt{\mathrm{GeV}}$ and a floor $b = 17.5$ ps [2]. That work also identified, as a priority for further study (its Section 7), a direct comparison of the timing performance of different wavelength-shifter (WLS) scintillator materials in otherwise-identical modules. This paper presents that comparison.

The central question is whether a lower-light WLS material such as LuAG:Ce times worse than the high-light organic DSB1 because of the species of scintillator — its emission spectrum, decay time, or refractive index — or simply because it collects less light. The distinction matters for detector design: if light yield is the sole driver, the timing of any candidate material is predictable from its light output, and the material can be chosen for radiation hardness or mechanical properties without a separate timing penalty. We answer it by measuring four builds spanning a factor ~ 2 in light yield with a common analysis, and by separating the light-dependent stochastic term from the irreducible floor.

A prerequisite is to time all builds on an equal footing despite saturation of the front end. We therefore first describe a high-gain/low-gain (HG/LG) ratio method that recovers the saturated rising edge (Section 3), then the estimator and selection (Section 4), and finally the results (Section 5): the four-build time resolution, its decomposition into a light-driven stochastic term and a material-independent floor, and the implications.

2. Experimental setup

2.1. Module and builds

The module is a tungsten/LYSO:Ce sampling calorimeter of the "shashlik" type: alternating plates of tungsten

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absorber and cerium-doped lutetium–yttrium oxyorthosilicate (LYSO:Ce) scintillator, ≈ 25 radiation lengths deep, read out longitudinally by WLS capillaries. Four capillaries traverse the stack and are each read out at both ends, giving $4 \times 2 = 8$ silicon-photomultiplier (SiPM) channels; we label the four downstream ends collectively DW and the four upstream ends UP. The transverse position of each incident electron is measured by a wire chamber and used to define a fiducial region on the beam spot.

Four builds were exposed, differing primarily in the WLS material of the capillaries – and hence in light yield – while sharing the same LYSO:Ce scintillator and tungsten absorber (Table 1): DSB1, high-light organic WLS; LUAG, low-light ceramic LuAG:Ce WLS; MIXED, a single module combining DSB1 and LuAG:Ce capillaries on opposite diagonals; and TENERGY, a multi-segment high-light build with an added longitudinal energy-readout section. DSB1, the brightest, is the build of Ref. [2].

2.2. Readout, beam, and reference

Each SiPM signal is split into a high-gain (HG) copy, optimised for timing, and a low-gain (LG) copy, optimised for energy linearity, both digitised by DRS4 switched-capacitor waveform digitisers (CAEN DT5742) [3] at 5 GS/s over a 1024-cell window, with per-cell calibration applied offline. The HG channels clip at ≈ 820 mV; the LG channels remain linear. Measurements used a 25–150 GeV electron beam at the CERN SPS (May 2023); DSB1 was run at all six energies in 25 GeV steps, the other builds at 50–150 GeV. A microchannel-plate photomultiplier (MCP-PMT) provides a common start; all per-channel times are MCP-referenced, and the differential estimator below cancels this reference.

3. Recovering the saturated edge: the HG/LG-ratio method

Precision timing of all builds on an equal footing requires the same well-defined edge in each, but the HG timing pulses saturate. The recorded HG amplitude piles up at the 820 mV clip (Fig. 1); for all but the smallest showers the waveform is flat across its maximum, and a constant-fraction discriminator (CFD) referenced to the measured peak is forced onto the shallow foot of the pulse, where the slope — and the timing information — is poor.

The LG copy, though too small for precise timing alone, is linear and therefore measures the true amplitude the HG channel would have reached. We use a per-end calibration

$$\mathrm{HG}_{\mathrm{true}} = a_{\mathrm{c}} + b_{\mathrm{c}}\,\mathrm{LG}_{\mathrm{peak}}, \tag{1}$$

fit on the unsaturated population (Fig. 2), to predict the unclipped HG peak event by event. A CFD threshold is then placed at a fixed fraction (15%) of this predicted peak and applied to the recorded HG waveform; because the predicted peak lies far above the clip, the threshold sits on

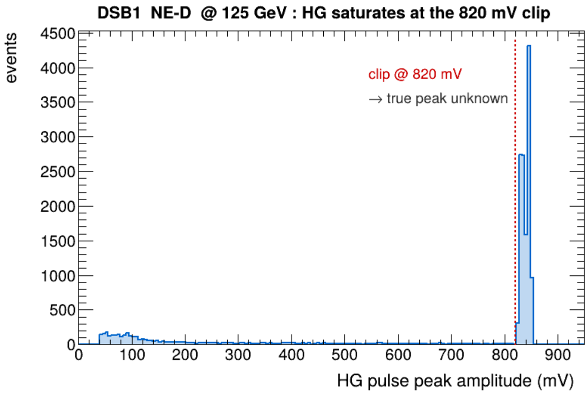

Figure 1: Distribution of recorded high-gain pulse-peak amplitudes (one channel, 125 GeV). The pile-up at ≈ 820 mV is the digitiser clip: most showers saturate, so the true peak — and the steep edge that leads to it — is not directly recorded.

the steep, unsaturated edge. We call the resulting MCP-referenced crossing time hg_lgcfd.

The recovered edge is a factor ~ 3.7 steeper than the clipped-peak foot (Fig. 3: 381 vs 102 mV/ns). Since the slew contribution to the time resolution scales as $\sigma_t \simeq \sigma_V/(\mathrm{d}V/\mathrm{d}t)$ with σ_V the voltage noise, a steeper edge directly improves timing. The method is applied uniformly to every build. For the low-light builds (LuAG, TENERGY), whose small pulses can defeat the fractional-threshold reconstruction, a robust fixed-threshold leading-edge time is used instead; the choice of per-build timing source is noted in Table 1.

4. Timing estimator and event selection

For each event we form the per-end mean crossing time and combine the two ends into the differential estimator

$$t_{\mathrm{DW-UP}} \equiv \tfrac{1}{2}(\bar{t}_{\mathrm{DW}} - \bar{t}_{\mathrm{UP}})\,, \tag{2}$$

averaging up to four downstream and four upstream SiPM times. The difference cancels any time common to both ends — in particular the MCP start — so $t_{\mathrm{DW-UP}}$ is free of the reference jitter; averaging four channels per end suppresses uncorrelated single-channel noise. The single-event resolution quoted throughout is the standard deviation of a Gaussian fit to the central peak of the $t_{\mathrm{DW-UP}}$ distribution; a small non-Gaussian shoulder (the ~ 0.2 ns satellite also noted in Ref. [2]) is excluded from the fit and has negligible effect on the quoted width.

Events are required to lie in the wire-chamber fiducial region and to have a valid MCP signal. Because shower fluctuations spread the collected light at fixed beam energy, the resolution depends on shower brightness; for a uniform, well-measured comparison across builds we quote the resolution of the 1000 brightest showers (by total low-gain signal) at each energy, which fixes the statistical precision of every point. Within each build, the resolution improves with shower brightness, as expected from the slew argument.

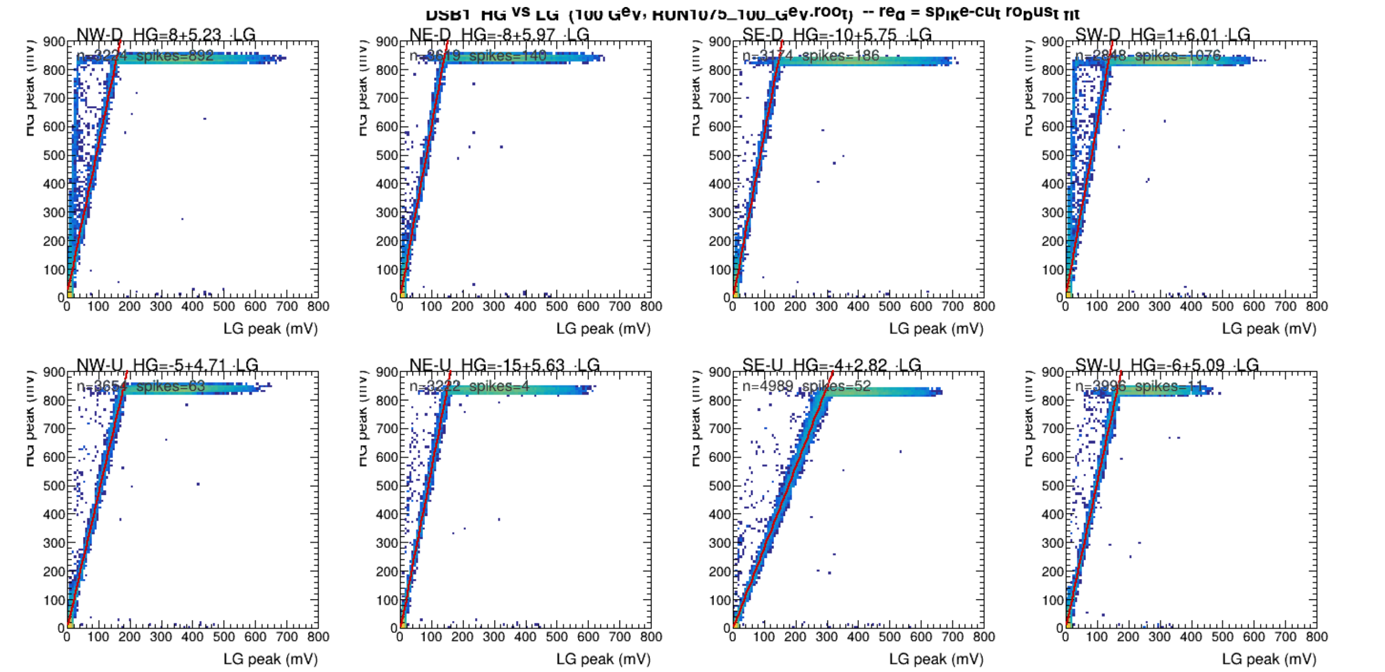

Figure 2: High-gain versus low-gain pulse-peak amplitude for the eight readout channels (100 GeV). The high-gain response saturates at 820 mV (horizontal band) while the low-gain copy remains linear; the per-channel linear fit (red line), Eq. (1), predicts the unclipped true peak.

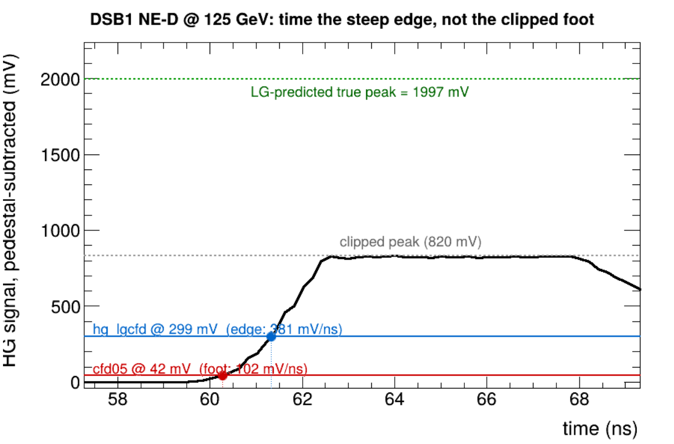

Figure 3: A single saturated high-gain waveform (125 GeV). The recorded pulse clips at 820 mV (grey); the low-gain copy predicts a true peak of 1997 mV (green). The CFD at 15% of the predicted peak (hg_lgcfd, blue) times an edge of 381 mV/ns, against 102 mV/ns at the clipped-peak foot where the 5% CFD (cfd05, red) must sit — a factor 3.7 steeper.

5. Results

5.1. Timing distributions

Figure 4 shows, for DSB1 with all energies pooled and binned by amplitude, the $t_{\mathrm{DW-UP}}$ (top) and absolute (DW+UP)/2 (middle) distributions with their central-peak Gaussian fits, and the fitted σ_t versus amplitude (bottom). The distributions are clean and single-peaked at every amplitude; the differential estimator is consistently narrower than the absolute one, as expected once the common-mode reference jitter is removed, and the width falls monotonically with shower brightness.

5.2. Light yield sets the resolution

Figure 5 is the central result. For each build, the brightest-shower σ_t is plotted against beam energy and fit to the published form

$$\sigma_t(E) = \frac{a}{\sqrt{E}} \oplus b, \tag{3}$$

with a the (light-driven) stochastic term and b the floor as $E \to \infty$. The fitted parameters are collected in Table 1. Two features stand out. First, the builds order cleanly by light yield, and the stochastic term tracks it: a rises from 201 ps $\sqrt{\mathrm{GeV}}$ for high-light DSB1 to 455 ps $\sqrt{\mathrm{GeV}}$ for low-light LuAG, a factor of 2.3. Second, the floor b is consistent across the LYSO-based builds — 20 ± 1 (DSB1), 22 ± 3 (TENERGY), 20 ± 6 ps (LuAG) — and agrees with the 17.5 ps reported for DSB1 in Ref. [2]. (MIXED sits higher, $b = 35 \pm 2$ ps; its low-gain readout is partly compromised, so its energy axis carries a systematic and its floor is indicative.) The energy dependence is well described by the $1/\sqrt{E}$ (photostatistics) form ($\chi^2/\mathrm{ndf} \approx 2$); a $1/E$ (pure-slew) form fits markedly worse.

The interpretation is direct: light yield sets the dominant, material-dependent (stochastic) part of the time resolution, while the floor is a shared, material-independent limit. A lower-light material times worse only in proportion to its reduced light; at sufficient light any of these materials approaches the same ~ 20 ps floor. The choice of WLS species per se is not a timing penalty.

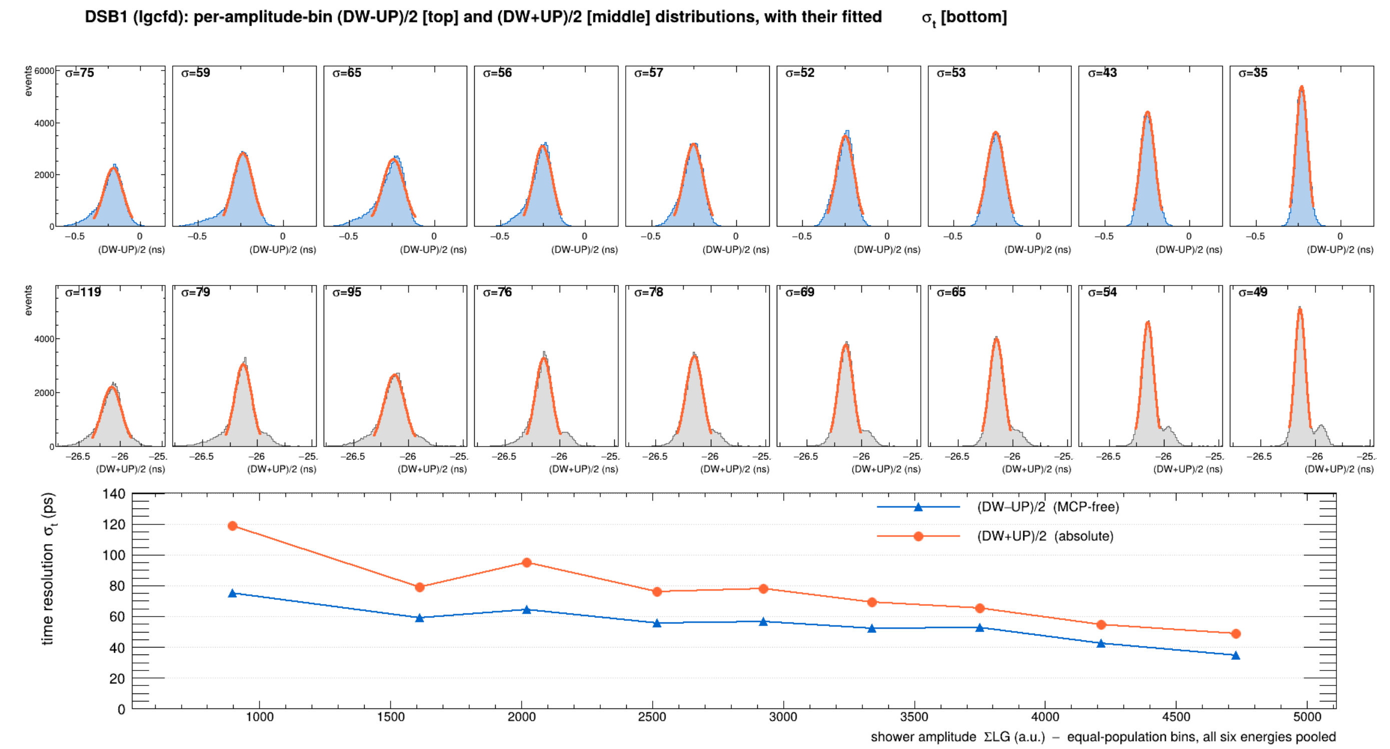

Figure 4: DSB1 timing distributions, all energies pooled and binned by amplitude. Top: (DW − UP)/2 (MCP-free). Middle: (DW + UP)/2 (absolute). Each σ (ps) is the central-peak Gaussian fit. Bottom: the fitted σ_t versus amplitude. The peaks are clean Gaussians at every brightness.

Table 1: Per-build fit of $\sigma_t = a/\sqrt{E} \oplus b$ to the brightest-shower resolution, the timing source used, and the best resolution measured at 150 GeV. The WLS column gives the wavelength-shifter material of the capillaries; the LYSO:Ce scintillator and tungsten absorber are common to every build. The published DSB1 values (cfd05) are shown for comparison.

Build	WLS	src	a	b (ps)	σ_t^{150}
DSB1	organic, hi	lgcfd	201	20 ± 1	25.4
TENERGY	organic, seg	led	240	22 ± 3	28.7
MIXED	org.+cer.	lgcfd	233	35 ± 2	40.0
LUAG	ceramic, lo	led	455	20 ± 6	41.2
DSB1 [2]	organic	cfd05	256	17.5	27

5.3. The gain from recovering the edge

The benefit of the HG/LG-ratio method is isolated by a direct comparison on identical showers (Fig. 6): the brightest-1000 DSB1 resolution is computed with cfd05 (a CFD on the clipped measured peak, as in Ref. [2]) and with hg_lgcfd (the recovered edge), the selection and estimator held fixed so that only the per-channel time differs. Timing the recovered edge improves the resolution by 3.1 ps at 150 GeV and lowers the floor from 22.1 ± 1.8 to 19.5 ± 1.1 ps. The gain is energy-dependent, largest where the timing is hardest: at high energy, where saturation is most severe, and at the lowest energy, where the 5% foot of a small pulse sits in noise; it is negligible at 50–75 GeV. The stochastic term is essentially unchanged (a = 205 → 201 ps $\sqrt{\text{GeV}}$), so the method acts chiefly on the high-energy/floor regime. (The cfd05 value here, a = 205, reflects the brightest-shower selection of this controlled comparison and differs from the 256 of Ref. [2] in Table 1, which used a different selection; holding the selection fixed isolates the timing method.) The best single-event resolution measured, for the brightest 150 GeV DSB1 showers, is 25.4 ps.

5.4. Systematic uncertainties

The headline brightest-1000 (DW − UP)/2 resolution was recomputed under deliberate variations of every selection knob, on the adopted timing source: the brightness selection (K = 500 and 2000 versus the nominal 1000), the timing fiducial radius (2.5 and 3.5 versus 3.0 mm), the MCP saturation window (upper cut 700 versus 750 mV), the per-channel HG threshold (30 versus 20 mV), and the fit energy range (50–150 versus 25–150 GeV). The shifts are collected in Table 2, with the total systematic taken as their RMS. The selection values themselves were fixed by the scans of Appendix Appendix A (Fig. A.7); the timing-method choice (cfd05 versus the recovered edge) is the subject of Sec. 5.3 and is not folded into this budget.

The headline is robust: the total selection systematic on $\sigma_t(150)$ is ≤ 1 ps for the high-light builds (DSB1, TENERGY) and ∼ 1.5 ps for MIXED and LUAG. The dominant single variation is loosening the brightness selection to K = 2000, which admits dimmer showers and genuinely

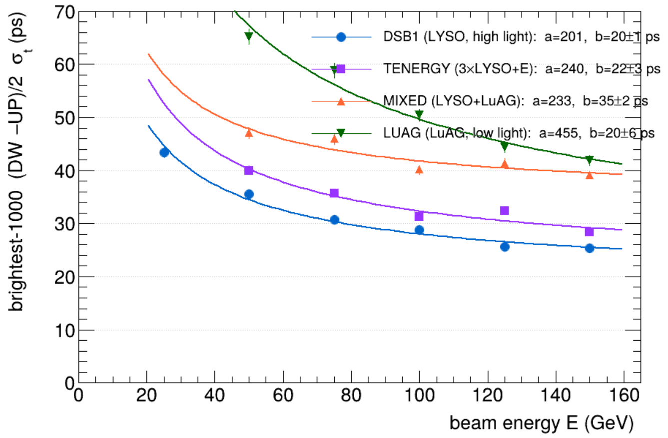

Figure 5: Brightest-shower (DW − UP)/2 time resolution versus beam energy for the four builds, each fit to $\sigma_t = a/\sqrt{E} \oplus b$ [Eq. (3)]. The stochastic term a tracks light yield (DSB1 lowest, LuAG highest); the floor $b \approx 20$ ps is consistent across the LYSO builds (MIXED, with a compromised low-gain readout, excepted).

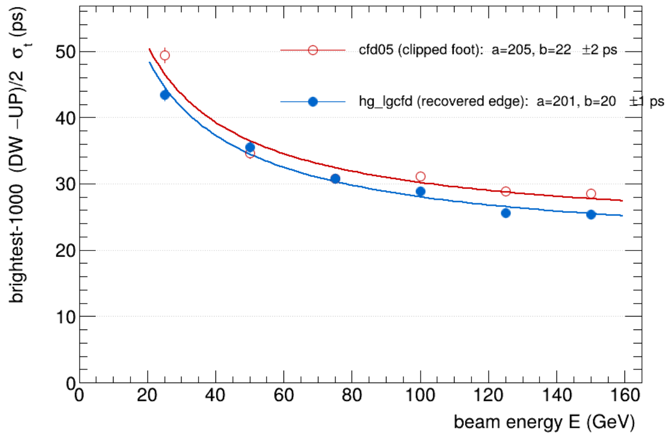

Figure 6: Time resolution of the brightest-1000 DSB1 showers timed with cfd05 (CFD on the clipped foot, open) and hg_lgcfd (CFD on the LG-recovered edge, filled), identical events and selection. Recovering the edge improves the resolution by ∼ 3 ps at 150 GeV and lowers the floor.

degrades the resolution by 2–2.4 ps — a manifestation of the same light–resolution dependence that is the paper's subject, not an analysis artefact. Crucially, the floor is stable: for DSB1, $b = 19.5 \pm 1.1\,(\mathrm{stat}) \pm 1.2\,(\mathrm{syst})$ ps, and across the LYSO builds b stays near 20 ps (TENERGY 21.6, LUAG 19.8) within the combined uncertainty, so the agreement with the published 17.5 ps survives the systematic variations. The elevated floor of the MIXED build (34.6 ps) is a distinct feature of that compromised configuration and carries the largest systematic, but does not affect the shared-floor conclusion drawn from the uniform builds.

6. Discussion

These measurements establish light yield as the primary lever for the timing of RADiCAL modules. Because the floor is material-independent, a WLS material may be selected for radiation hardness, mechanical robustness, or cost, with its timing then set by — and predictable from — its light output. LuAG:Ce, attractive for its radiation hardness [4], times worse than DSB1 by exactly the factor its lower light yield implies, not more.

The floor $b \approx 20$ ps is reached, within errors, by all the LYSO builds and matches the previously published value. Its origin is plausibly the shower itself: the (DW − UP)/2 estimator is, by construction, sensitive to the longitudinal position of energy deposition (light from a deeper shower reaches the two ends with opposite timing shifts), so shower-to-shower depth fluctuation produces an irreducible spread that does not decrease with light and is common to any scintillator filling the same geometry. This same depth sensitivity suggests the route below the floor: a depth-corrected estimator, using the channel-to-channel light-sharing to estimate and subtract each shower's depth, which we are pursuing.

7. Conclusions

Comparing four WLS builds of a tungsten/LYSO:Ce sampling calorimeter on a common footing — enabled by a high-gain/low-gain-ratio method that recovers the saturated rising edge — we find that the time resolution is governed by scintillation light yield rather than wavelength-shifter species. The stochastic term of $\sigma_t = a/\sqrt{E} \oplus b$ more than doubles from the high-light to the low-light build, while the floor $b \approx 20$ ps is consistent across the LYSO builds and with the previously published DSB1 value, confirming and extending that result to multiple materials. The high-light build reaches 25.4 ps for the brightest 150 GeV showers. Light yield is therefore the design lever for fast-timing calorimetry with these modules, and the residual floor — consistent with shower-depth fluctuation — is the target of a depth-corrected analysis to follow.

Appendix A. Selection optimization and systematic stability

The four choices that define the headline estimator were each fixed by a scan at the headline energy (150 GeV), shown for all builds in Fig. A.7. Panel (a) compares timing sources: a constant-fraction time on the clipped peak, swept over fractions 3–50% (solid), against the recovered-edge source we adopt (lgcfd/led, dashed). The adopted source lies below the entire clipped-peak family for every build — dramatically so for the low-light LUAG — which is the quantitative basis for the method of Sec. 3. Panel (b) shows the brightness selection: σ_t falls steeply as the dimmest showers are removed and flattens near

Table 2: Systematic budget for the brightest-1000 (DW − UP)/2 time resolution. Each row recomputes $\sigma_t(150)$ and the fit floor b under one selection variation; the entry is the shift from the nominal value (ps). The total systematic is the RMS of the shifts; the statistical floor error is from the fit. The timing-method choice (cfd05 vs. the recovered edge) is the subject of Sec. 5.3 and is not folded in here.

Variation	DSB1	TENERGY	MIXED	LUAG
shift in $\sigma_t(150)$ [ps]				
K=500	+0.1	−0.3	−1.0	+0.8
K=2000	+2.1	+2.4	+2.3	+0.4
r_{fid}=2.5 mm	+1.2	+0.3	−0.7	−1.1
r_{fid}=3.5 mm	+0.2	+0.6	+2.0	+3.8
MCP < 700 mV	+0.3	+0.2	+2.5	−0.7
HG > 30 mV	+0.0	−0.0	+0.0	+0.0
fit 50–150 only	+0.0	+0.0	+0.0	+0.0
nominal $\sigma_t(150)$	25.3	28.3	39.1	41.9
total syst.	±0.9	±0.9	±1.5	±1.6
floor b (nom)	19.5	21.6	34.6	19.8
b syst.	±1.2	±2.2	±2.2	±2.4
b stat.	±1.1	±2.7	±2.4	±5.6

$K \sim 1000$, beyond which admitting dimmer showers degrades it again; $K = 1000$ sits at the knee. Panel (c) shows the resolution is flat across the timing fiducial radius out to the chosen 3.0 mm, rising only as off-axis showers leak in. Panel (d) shows insensitivity to the per-channel HG threshold, confirming the 20 mV working point is well clear of the noise floor. The corresponding systematic stability of $\sigma_t(150)$ under these variations is summarised in Fig. A.8 and Table 2.

Acknowledgements

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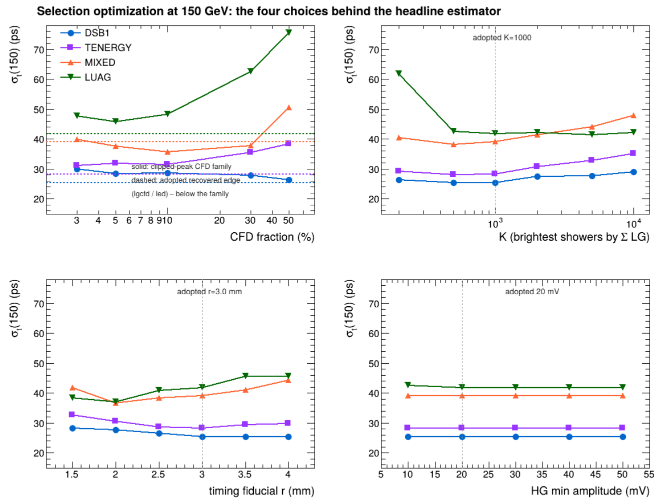

Figure A.7: Selection optimization at 150 GeV for all four builds: (a) timing source as a function of CFD fraction on the clipped peak (solid) with the adopted recovered-edge source overlaid (dashed); (b) resolution versus the brightest-N selection K; (c) versus timing fiducial radius; (d) versus the per-channel HG threshold. Dashed grey lines mark the adopted working points.

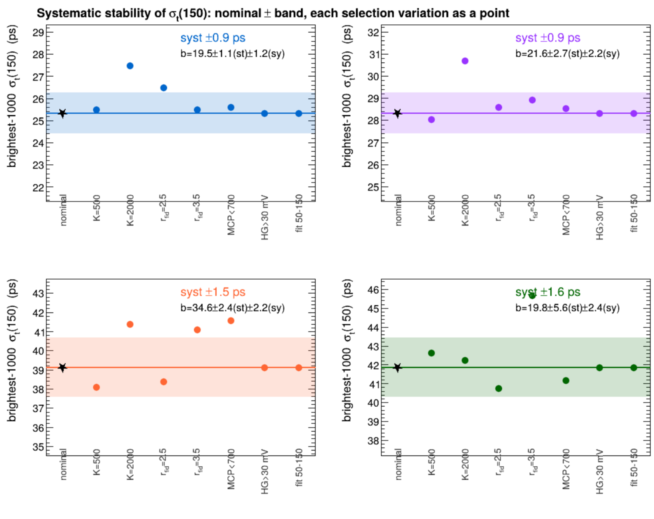

Figure A.8: Systematic stability of the brightest-1000 $\sigma_t(150)$ for each build: the nominal value (star) with its total-systematic band, and each selection variation as a point. The fitted floor b with its statistical and systematic errors is annotated per build. All variations lie within the band except the deliberate $K = 2000$ loosening of the brightness selection.